Problem set 2

Due 10/12/26

1 Iterated deletion

The problem is based on the game matrix below.

		Phil		
		n_P	l_P	h_P
	n_G	$0,0$	$1,2$	$1,2$
Greg	l_G	$3,1$	$3,3$	$3,4$
	h_G	$2,1$	$4,3$	$5,5$

1.A). Find the solution set of strategies based on iterated deletion of strictly dominated strategies.

1.B). Find the set of Nash equilibrium.

2 Putting games into the Normal Form

Consider the following 2 stage game $t = 1, 2, 3$. The game has 2 players. In stage 1 player 1 takes one of two $\{a_1, b_1\}$. In stage 2 player two first observes that action of player 1 in stage 1 and then takes one of three actions $\{c_2, d_2, e_2\}$. The payoff of each player is as follows.

$$\begin{array}{cc} u_1(a_1, c_2) = 10 & u_2(a_1, c_2) = 8 \\ u_1(a_1, d_2) = 8 & u_2(a_1, d_2) = 9 \\ u_1(a_1, e_2) = 12 & u_2(a_1, e_2) = -11 \\ u_1(b_1, c_2) = 13 & u_2(b_1, c_2) = 0 \\ u_1(b_1, d_2) = 1 & u_2(b_1, d_2) = 10 \\ u_1(b_1, e_2) = 10 & u_2(b_1, e_2) = 6 \end{array}$$

1. Draw the game tree of the game.
2. Write out the set of strategies for each player.
3. Draw the normal form matrix for the game.

4. Find all Nash equilibrium.

5. Are there equilibrium that are based on incredible threats? Explain

3 Timing and Strategies

Take a simple two player game with each player having actions 0 or 1. Player 1 gets \$1 if the actions match and \$0 if they do not match. Player 2 gets \$0 if the actions match and \$1 if they do not match.

2.1 Suppose the game is simultaneous choice game. Explain the strategies of each player.

2.2 Suppose the game is sequential choice game in which player 1 goes first, then player 2 observes player 1's choice and makes her choice. Explain the strategies of each player.

2.3 Explain the difference in the normal form of these two games.

4 Third party observation of a game

Suppose that you run an experiment in a laboratory with real money in which there are many pairs of players and the game has the following structure.

Each player i has three actions $\{L_i, M_i, H_i\}$. The money each player i receives for the different action pairs is (in a game with player j) is described in the matrix below:

	L_1	M_1	H_1
L_2	\$8, \$8	\$30, \$10	\$15, \$1
M_2	\$10, \$30	\$18, \$18	\$20, \$4
H_2	\$1, \$15	\$4, \$20	\$5, \$5

Your experiment is a simultaneous game with each player picking their actions simultaneously and independently. You observe the outcomes of the games and ever player i in the sample pick the action M_i.

2.4 Does the observed behavior reject a game theoretic explanation? Explain your answer.

2.5 Clearly define and explain the idea of a normal form of a game.

2.6 Postulate an explanation for the behavior observed in the experiment and create a normal form of this game with the unique Nash equilibrium of $\{M_1, M_2\}$.

Can we use empirical data from an experiment of this form to test if the players are rational? Explain.

5 Subgame Perfect Nash Equilibrium

Consider the following sequential game. Player 1 acts first and picks an action in the set $\{L_1, M_1, H_1\}$ and then after observing player 2's action, Player 1 takes an action in $\{U_2, D_2\}$. The payoffs if the players are as follows:

$$\begin{array}{ll} u_1(L_1, U_2) = 10 & u_2(L_1, U_2) = 10 \\ u_1(L_1, D_2) = 1 & u_2(L_1, D_2) = 11 \\ u_1(M_1, U_2) = 5 & u_2(M_1, U_2) = 5 \\ u_1(M_1, D_2) = 2 & u_2(M_1, D_2) = 6 \\ u_1(H_1, U_2) = 1 & u_2(H_1, U_2) = 4 \\ u_1(H_1, D_2) = 3 & u_2(H_1, D_2) = 5 \end{array}$$

1. Find the Normal Form of the game.
2. Draw the Normal From game matrix and find all Nah Equilibrium
3. Draw the Game Tree for this game and find all Subgame Perfect Nash Equilibrium.